

JP24July

Online Friendly Challenge 2024

Mobile Applications Development (Skill 08)

Module C Implementation

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Introduction

SoundScene is an innovative tablet application designed to plan and simulate sound sources within a given room. By accurately modeling how sound propagates and interacts with the environment, SoundScene aims to enhance auditory orientation for visually impaired individuals, ultimately improving their quality of life.

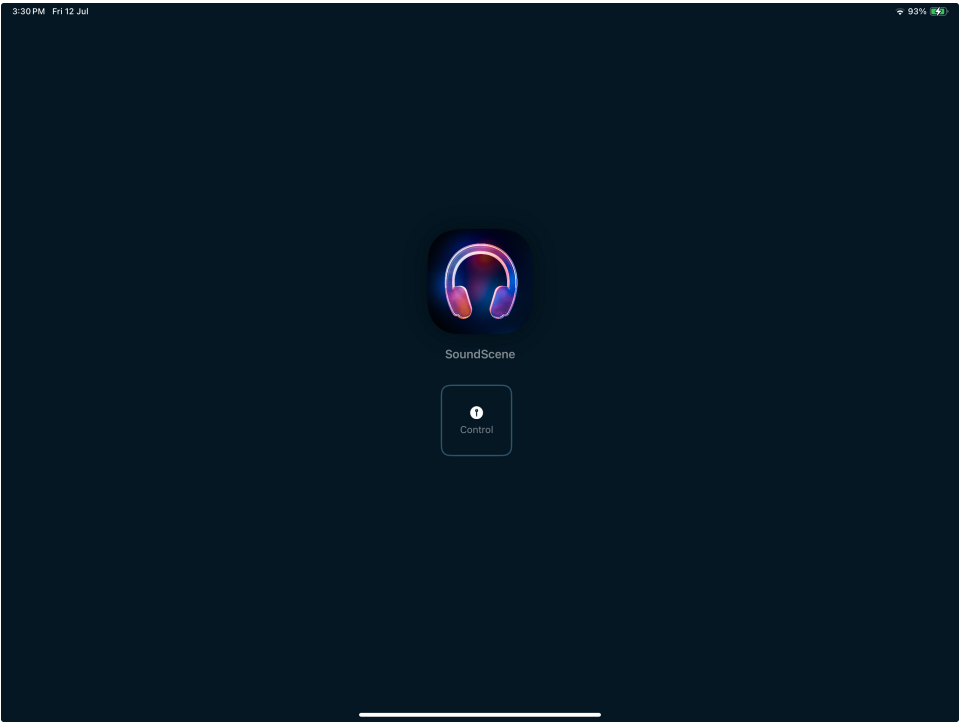
The designer of the **SoundScene** team has provided you with the high-fidelity design. As a mobile application developer, your task is to strictly follow the design guide to implement the user interface and interactivity of the app, and also implement a few functionalities with respect to the interaction design.

Description of project and tasks

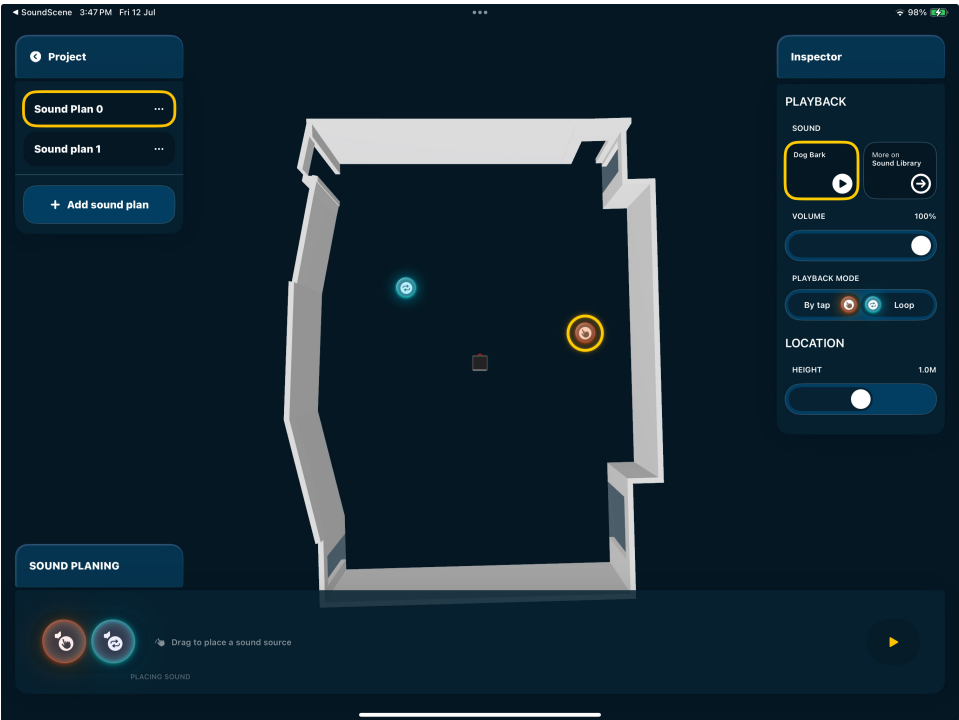
Module	Name	Competition Time	Marking Devices
A	Implementation (Tablet)	2.5h	Emulator : iOS: iPad 11 Pro (iOS 16.4) Android: Pixel C (API level: 31)

Below is an overview of some wireframes and functionalities. Please refer to the files/ folder for detailed information.

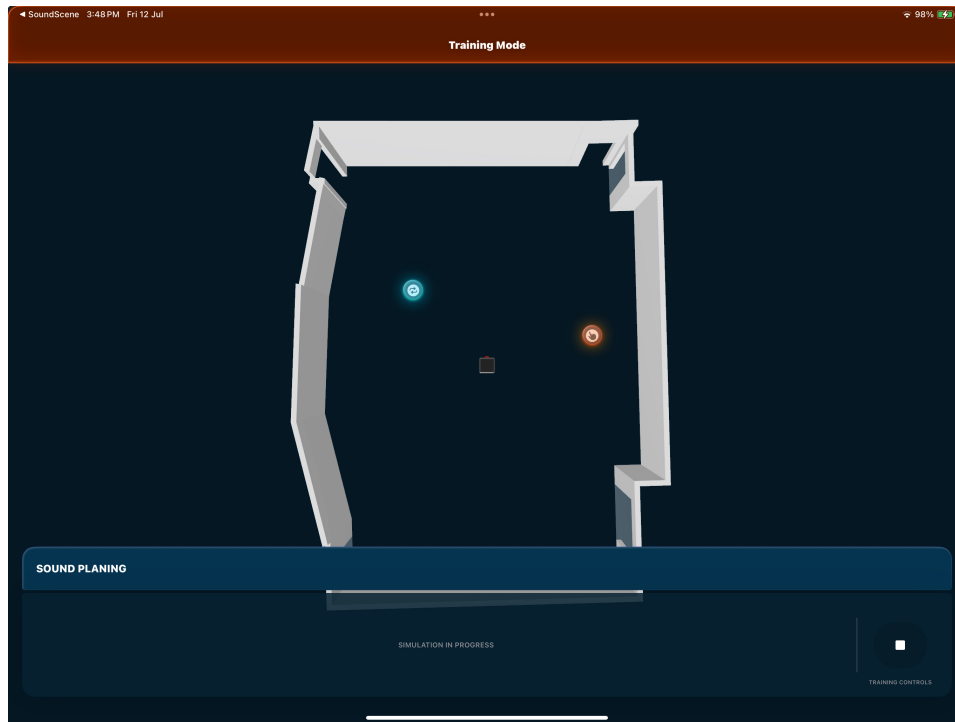
Application Design



Landing Page



Editing Page (Planning Mode)



Editing Page (Simulation Mode)

General Demands [2.0]

1. Complete the Proof of Concept app according to the given specifications.
2. You don't need to strictly follow the wordings shown on the given design. However, please make sure the corresponding UI element could be identified easily by the test user.
3. The package name / bundle identifier is expected to contain "xx.com.jp24ofc", xx means your country code in small letter. (e.g., China: cn, Japan: jp)
4. You don't need to make a 100% replica of the design, but you are required to complete the app in a high level of similarity in terms of the UI elements positioning, the use of colour, the use of icons and animations.
5. The application should show the application icon (media-files/image/applcon.png) on the home screen. [0.1]
6. The app should be forced in landscape orientation (no restriction for left / right landscape) [0.1]
7. When the app launches, the app should showing the Landing Page. [0.1]
8. You should design the data model for the sound plan. The sound plan could be saved stored locally (in any form) and could be displayed properly after restarting the app. [0.5 (1.0: this feature is partially implemented with/without save), 1.5: full implementation)]

Suggested structure of the data model (not compulsory for you to fully adopt this structure):

Sound plans:

Fields	Type
id	String , Unique identifier
name	String
soundSources	An Array of Sound Source

Sound Source:

Fields	Type
id	String , Unique identifier
name	String
kind	String: "ByTap" or "Loop"
position	Structure with x and y values in double type
entryId	ID of the sound effect (refer to sound library)

9. Submit your work to the given git repositories. [0.2]

Pages Demands

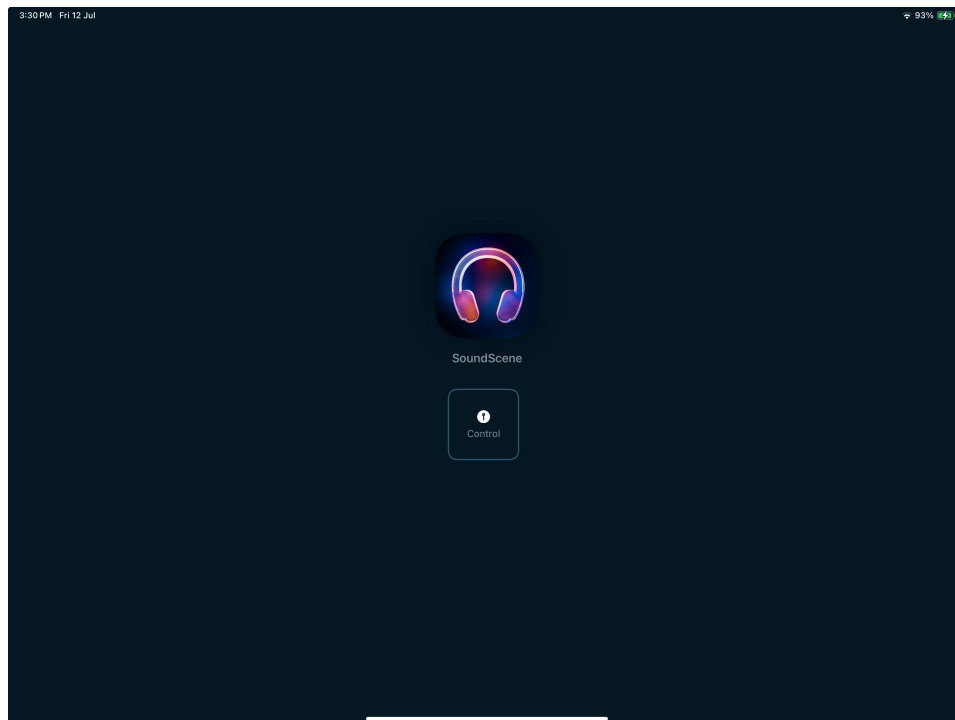
1. Landing Page

1.1 Elements & Appearance include: [0.6]

1. App Icon using the given icon image (media-files/image/applcon.png) [0.1]
2. "SoundScene" Label [0.1]
3. "Control" Button, which includes
 - 3.1 "Control" Label [0.1]
 - 3.2 An icon [0.1]
 - 3.3 A border [0.1]
4. Background in dark colour [0.1]

1.2 Functional requirements: [0.2]

1. When the user clicks the "Control" button, the app will shows the "Editing Page" with ANY transition animation. [0.2]



2. Editing Page

2.1 Elements & Appearance include: [0.6]

1. Project Panel [0.1]
2. Inspector Panel (Sound Library Panel would be inside this panel as well) [0.1]
3. Bottom Panel [0.1]
4. Sound Map [0.1]
5. "Training Mode" [0.1] title bar with decoration based on given design. [0.1]

2.2 Functional requirements: [1.0]

This page contains two mode, the "EDITING" mode and the "SIMULATION" mode. The app can switch between "EDITING" mode and "SIMULATION" mode by pressing the Start / Stop button at the Bottom Panel (Task 7). There are two kind of sound source: "By Tap" and "Loop". The sound sources will be placed on the Sound Map (Refer to Task 6 & Task 7).

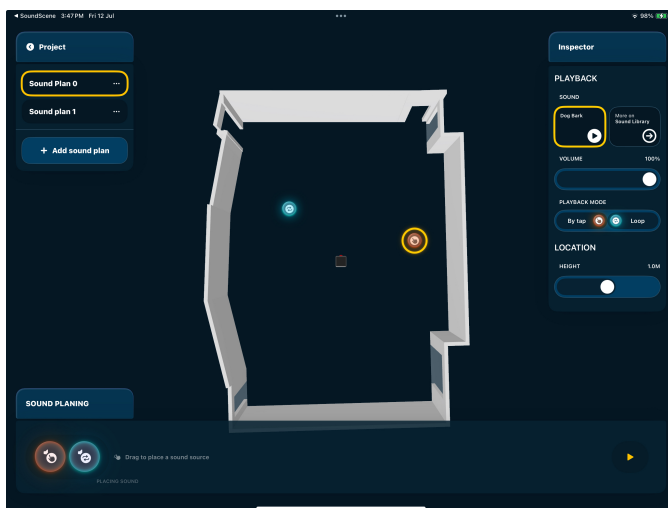
1. You are required to implement the animation of switching. ([media-files/animation/switching.mp4](#)) [0.2]

In Editing Mode:

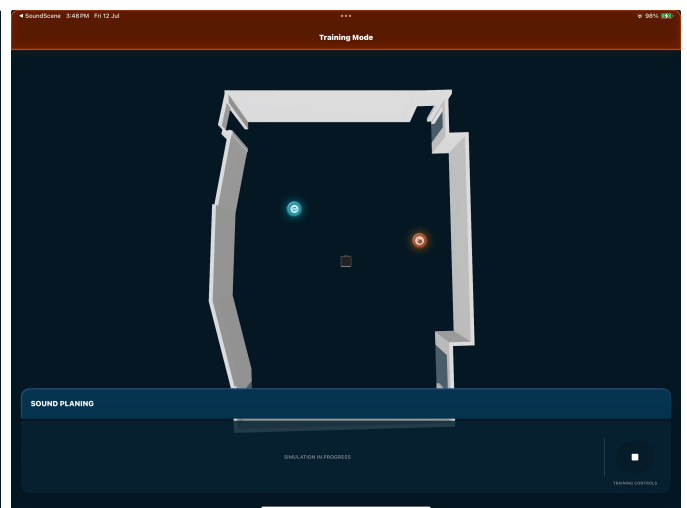
- a. When the user tap on the sound source on the sound map, the sound source will be selected with highlight. [0.1]

In Simulation Mode:

- b. When the user enters this mode, all "Loop" sound sources will play automatically [0.2] with animations. ([media-files/animation/ripple.mp4](#))
- c. The the user enters this mode, selected sound source will be deselected. [0.1]
- d. When the user tap on the "By Tap" sound source, the tapped sound source will play / pause sound respectively [0.2] with animation. ([media-files/animation/ripple.mp4](#))
- e. When the user leave this mode , all sound sources will stop. [0.2]



Editing Page (Plan Mode)

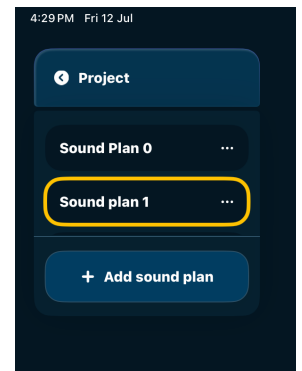


Editing Page (Simulation Mode)

3. Editing Page - Project Panel

3.1 Elements include: [0.5]

1. Navigation Bar
 - 1.1 "Sound library" title [0.1] with icon indicate the back action [0.1]
2. Content Area
 - 2.1 Sound Plan Listing. [0.1]
 - 2.2 "Sound Plan Cell with a sound plan name label. [0.1]
 - 2.4 A button "Add sound plan" label with icon [0.1].



3.2 Functional requirements: [0.9]

1. When the page shows, the first sound plan will be selected and the sound map (task 6) will also show the stored sound sources. For first launch, please create and show a empty sound plan. [0.2]
2. When the user clicks the "Project" label, the app will go back to landing page. [0.1]
3. All playing sound sources/effects will be paused. [0.1]
4. When the user clicks the "Add sound plan", a new empty sound plan will be shown in the listing. [0.2]
5. When the user clicks on unselected sound plan, the particular will be selected with highlight. (single select only) [0.1]
6. When a sound plan is selected, the sound map (Task 6) will update with animation. [0.2] ([media-files/animation/changeplan.mp4](#))

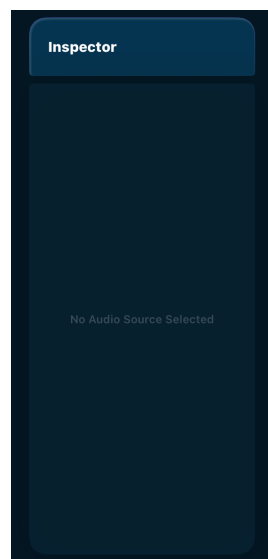
4. Editing Page - Inspector Panel

4.1 Elements include: [1.2]

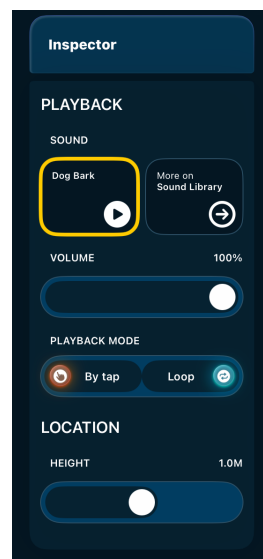
1. Navigation Bar
 - 1.1 "Inspector" label [0.1]
2. Content Area
 - 2.1 "PLAYBACK" label [0.1]
 - 2.2 Selected sound cell [0.1] (Default sound: dog bark)
 - 2.3 More on sound library cell with enter button [0.1]
 - 2.4 "VOLUME", volume label and Percentage slider (range: 0.0 ~ 1.0, default: 1.0) [0.1]
 - 2.6 "PLAYBACK MODE" label [0.1]
 - 2.7 "By Tap" switch [0.1] and "Loop" switch [0.1], decorated with given design [0.2]
 - 2.8 "LOCATION" label [0.1]
 - 2.9 "HEIGHT", height value label and height slider (range: 0.0 ~ 2.0, default: 0.5) [0.1]

4.2 Functional requirements: [1.6]

1. When no sound source selected, the inspector panel will show "No Audio Source Selected" label [0.1].
2. When a sound source is selected, the inspector panel correctly shows the value of the corresponding source code [0.2].
3. The inspector will change the stored value of the selected sound source, which can be reflected in next selection [0.2]
4. Each sound source should be either "By Lap" or "Loop", this feature should be reflected in the switches [0.2]
5. When the user clicks the enter button of "More Sound Library", the inspector will change to "Sound Library" with animation. [0.2]
6. When the user start changing the value of the height, it will blur the sound plan and unselected sound sources. [0.2] A height indicator in human shape will be shown and move with respect to the height value. (*Reference: media-files/animation/height.mp4*) [0.3]
7. The sound can be played and paused, by pressing the play button at the selected sound cell. You are required to play ripple animation at the selected sound source UI. (Task reference to task 7). [0.2]



Inspector (No selection)



Inspector (With selection)

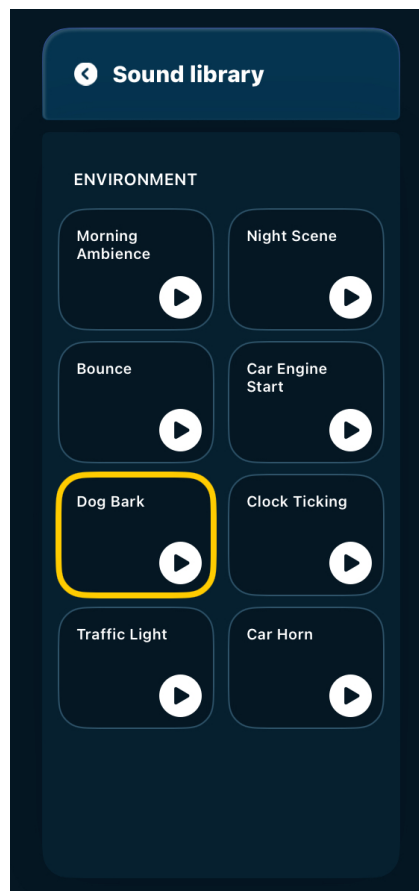
5. Editor Page - Sound Library Panel

5.1 Elements include: [0.6]

1. Navigation Bar
 - 1.1 "Sound library" title with an icon indicates the back action [0.1]
2. Content Area
 - 2.1 "Environment" Label [0.1]
 - 2.2 Sound Library Grid View [0.1]
 - 2.2 "Sound Library Cell" decorated with border [0.1]. Inside the cell, there are sound name label [0.1] and play button inside. [0.1]

5.2 Functional requirements: [1.3]

1. The sound library grid view shows a list of available sound effects. (*media-files/sound/library.json*) [0.3]
2. When the user clicks the "Sound Library" title, it will go back to inspector [0.1] of the selected sound source. All playing sound source will be paused. [0.2]
3. When the user clicks the play button of a sound library cell, it will play the corresponding sound (can be located at the *media-files/sound/* folder) [0.2] with appropriate UI change. The sound can be played simultaneously with other sound source. [0.1]
4. When the user clicks pause button of a playing source, it will pause the playing sound with appropriate UI change. [0.1]
5. When the user click on a sound source, the corresponding sound cell will be highlighted. When the user goes back to inspector view, it will reflect the change. [0.2]
6. When the sound playback ended, the pause button will revert to play button. [0.1]



6. Editor Page - Bottom Panel

6.1 Elements include: [1.1]

1. Title Bar
 - 1.1 "SOUND PLANING" label [0.1]

2. Content Area

In Editing Mode:

- 2.1 A "By Tap" Sound Source [0.1], decorated based on given design [0.1]
- 2.2 A "Loop" Sound Source [0.1], decorated based on given design [0.1]
- 2.3 A "Drag to place a sound source" Label with icon [0.1]
- 2.4 A "PLACING SOUND" label [0.1]
- 2.5 A Play Button [0.1]

In Simulation Mode:

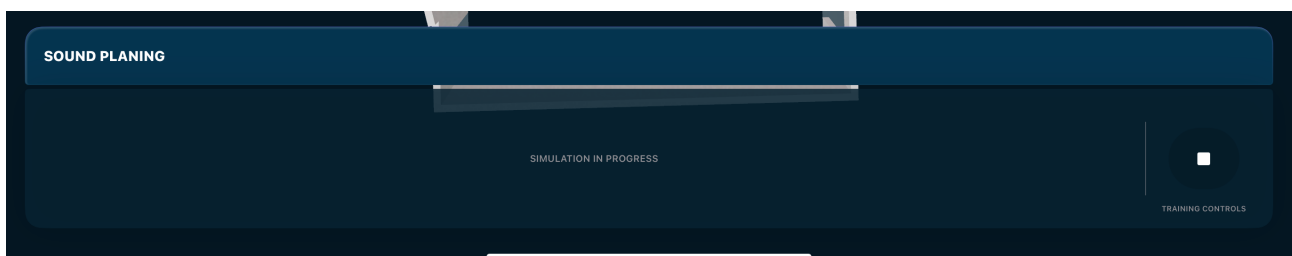
- 2.6 "SIMULATION IN PROGRESS" label [0.1]
- 2.7 A Separator [0.1]
- 2.8 A Stop Button [0.1]

6.2 Functional requirements: [1.0]

1. Switch the app into Editing / Simulation mode by pressing the Start/Stop Button [0.1]
2. During the switching, the Title Bar resizes with animations. [0.1]
3. The "By Tap" and "Loop" Sound Source can be dragged. [0.4, (0.2 for each view)]
4. The "By Tap" and "Loop" will go back to the original place with **ANIMATION** with reference to the suggested animation. ([animation/gobackandpopup.mp4](#)) [0.4, (0.2 for each view)]



Bottom Panel (Editing Mode)



Bottom Panel (Simulation Mode)

7. Editor Page - Sound Map

7.1 Elements include: [0.3]

1. A background using the image as the map of a room [0.1] ([media-files/image/map.png](#))
2. Icons represented as "Sound Source" on the map. [0.1], decorated based on design [0.1].

7.2 Functional requirements: [1.2]

1. The sound source can be selected [0.1] and a highlight is shown. [0.1]
2. All sound sources can be dragged[0.1] and relocated
3. Sound source will show animation while the sound is play. [0.5] ([Reference: media-files/animation/ripple.mp4](#))
4. When the sound map receives dragged sound plan from the Bottom Panel (Task 6), a new instance of the sound source will be created and the new sound source will pop out with animation ([Reference: media-files/animation/gobackandpopup.mp4](#)). [0.4]

Instructions to the Competitor

1. You need to compress the dedicated files and related resource files into XX_Module_A.zip, export the mobile application as XX_Module_A.apk (for Android) or XX_Module_A.ipa/XX_Module_A.app (for iOS) executable files, and finally place them into the XX_Module_A folder.
2. The entire XX_Module_A folder should be pushed to the provided remote Git repository.

Note: XX refers to your country code.

Marking Scheme

Module A		
No.	Sub-criterion	Mark
1	Landing Page	0.6
2	Editor Page	1.6
3	Editor Page - Project Panel	1.4
4	Editor Page - Inspector Panel	2.8
5	Editor Page - Sound Library	1.9
6	Editor Page - Bottom Panel	2.1
7	Editor Page - Sound Map	1.5
8	General Demands	2.0
9	Judgement	2.5
	Total	16.4

Judgement (2.5 marks):

Compliance to the given design. (1.5 marks)

0 - many UI elements / pages are missing.

1 - the app somehow works but inappropriate UI / app navigation is implemented.

2 - most of the UI / app navigation and UI appearance follow the design with some missing elements

3 - perfectly or nearly perfectly follow the high fidelity design

The quality of implementation of functionalities and animations. (1.0 marks)

0 - the app cannot runs / comes with very limited functionalities/animations.

1 - the app somehow offers limited functionalities / animations.

2 - most of the functionalities an animations are implemented with several problems / missing elements.

3 - Almost all of the functionalities are implemented.